

## FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

### FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.1 TURNING EFFECT OF FORCE: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

#### INSTRUCTIONS GIVEN TO STUDENTS

Think back to the Jembe Handle Puzzle from Lesson 1. In Kericho, Amina used a long-handled jembe (about 90 cm) and finished digging quickly and easily. Juma used a jembe with the same blade weight but a short handle (about 30 cm), and he struggled and tired quickly. Your task is to explain WHY, using the evidence you collected in this unit. Write your answers in the space under each prompt. Use full sentences, show every step of any calculation with units, and use the words pivot, force, perpendicular distance, moment, clockwise and anticlockwise correctly. Use  $g = 10 \text{ N/kg}$  where you need weight. Write in your own words, as if explaining to a farmer in your home area.

#### Part 1: Finding the pattern in the metre-rule data

In the metre-rule investigation, a wooden metre rule was balanced on a pencil pivot at the 50 cm mark. Slotted masses were hung on each side. Here is the data from one group. Every trial was balanced. Use  $W = mg$  with  $g = 10 \text{ N/kg}$ .

| Trial | Left mass (g) | Left distance from pivot (m) | Right mass (g) | Right distance from pivot (m) |
|-------|---------------|------------------------------|----------------|-------------------------------|
| 1     | 500           | 0.10                         | 200            | 0.25                          |
| 2     | 10            | 0.4                          | 20             | 0.2                           |

(a)  $100 \text{ g} = 1 \text{ N}$ ,  $200 \text{ g} = 2 \text{ N}$ ,  $500 \text{ g} = 5 \text{ N}$ .

(b) Trial 1: left  $5 \text{ N} \times 0.10 \text{ m} = 0.5 \text{ N}\cdot\text{m}$ ; right  $2 \text{ N} \times 0.25 \text{ m} = 0.5 \text{ N}\cdot\text{m}$ . Trial 2: left  $1 \text{ N} \times 0.40 \text{ m} = 0.4 \text{ N}\cdot\text{m}$ ; right  $2 \text{ N} \times 0.20 \text{ m} = 0.4 \text{ N}\cdot\text{m}$ . Trial 3: left  $2 \text{ N} \times 0.15 \text{ m} = 0.30 \text{ N}\cdot\text{m}$ ; right  $1 \text{ N} \times 0.30 \text{ m} = 0.30 \text{ N}\cdot\text{m}$ .

(c) In every trial the left  $F \times d$  equals the right  $F \times d$ . This is the Principle of Moments: a body is in rotational equilibrium (balanced) when the sum of the clockwise moments about the pivot equals the sum of the anticlockwise moments about the pivot. The moment of a force is  $M = F \times d$ , measured in  $\text{N}\cdot\text{m}$ .

(d) The rule balances because what matters is the moment, not the force or the distance alone. For example, in Trial 2 the  $1 \text{ N}$  force at  $0.40 \text{ m}$  gives the same  $0.4 \text{ N}\cdot\text{m}$  turning effect as the  $2 \text{ N}$  force at  $0.20 \text{ m}$ . The turning effects on the two sides cancel, so the rule does not rotate.

|   |         |          |         |          |
|---|---------|----------|---------|----------|
|   | 0       | 0        | 0       | 0        |
| 3 | 20<br>0 | 0.1<br>5 | 10<br>0 | 0.3<br>0 |

(a) Convert each mass to a weight (force) in Newtons. (b) Calculate  $F \times d$  for the left side and for the right side in each trial, with units. (c) What pattern do you find? State it as a general rule in words and give it its scientific name. (d) Why is the metre rule balanced even though the masses and the distances on the two sides are different?

## Part 2: Explaining the Jembe Handle Puzzle

**Amina (handle about 90 cm) and Juma (handle about 30 cm) used jembes with identical blade weights to dig the same soil. (a) In the jembe, which point acts as the pivot, and what is the distance  $d$ ? (b) Assuming each of them applies the same effort force, use the equation  $M = F \times d$  to explain why Amina found the digging easier than Juma. (c) By roughly how many times is Amina's handle length greater than Juma's? Use this ratio to compare the moments they produce. (d) A classmate says: 'Amina must have pushed with a much bigger force.' Use the idea of moments to respond to this claim.**

- (a) The pivot is the point where the blade meets the soil. The distance  $d$  is the perpendicular distance from the pivot to the line along which the farmer's effort acts, which is measured along the handle to the grip.
- (b) The moment is  $M = F \times d$ . Amina's handle is longer, so her grip is further from the pivot and her  $d$  is larger. With the same effort force  $F$ , her larger  $d$  gives a larger moment, so a greater turning effect on the blade. The soil is therefore broken with less effort from her. Juma's short handle gives a small  $d$ , so the same force produces a small moment. He must work much harder to get a similar turning effect, and he tires quickly.
- (c)  $90 \text{ cm} \div 30 \text{ cm} = 3$ , so Amina's handle is about 3 times longer. For the same force her moment is about 3 times greater than Juma's, because the moment is directly proportional to  $d$  when  $F$  is unchanged.
- (d) A larger moment can come from a larger force OR a larger distance. Amina's bigger moment comes from her larger perpendicular distance  $d$ , not from a larger  $F$ . Since  $M = F \times d$ , increasing  $d$  increases the turning effect without pushing harder.

### Part 3: Calculating moments and the net turning effect

Look at this version of the jembe problem. Treat the jembe as able to turn about a fixed pivot P. A farmer grips the jembe 0.8 m from P and pushes with a force of 50 N, which turns the jembe anticlockwise about P. Friction at the blade applies a force of 20 N at a distance of 0.1 m from P, on the other side of P, which turns the jembe clockwise. (a) Calculate the anticlockwise moment and the clockwise moment, showing the equation and units. (b) Calculate the net moment. (c) Will the jembe rotate? If so, in which direction? (d) Why must you give each moment a direction (clockwise or anticlockwise) before you compare them? (e) When you work out a moment from a diagram, which distance must you use for d, and what common mistake do learners make when choosing it?

- (a) Anticlockwise moment =  $F \times d = 50 \text{ N} \times 0.8 \text{ m} = 40 \text{ N}\cdot\text{m}$ . Clockwise moment =  $20 \text{ N} \times 0.1 \text{ m} = 2 \text{ N}\cdot\text{m}$ .
- (b) Net moment =  $40 \text{ N}\cdot\text{m} - 2 \text{ N}\cdot\text{m} = 38 \text{ N}\cdot\text{m}$  in the anticlockwise direction.
- (c) Yes. The anticlockwise moment is larger than the clockwise moment, so the jembe rotates anticlockwise about P.
- (d) Moments that turn in opposite directions work against each other, so they must be subtracted, not added. Without directions we cannot tell which way the object turns, or whether the moments balance. Equilibrium needs the clockwise moments to equal the anticlockwise moments.
- (e) d must be the perpendicular distance from the pivot to the line of action of the force. A common mistake is to use the length of the force arrow instead. The arrow's length only shows the size of the force and is not the distance d.

### Part 4: Two hands on a steering wheel – couples

A matatu driver uses TWO hands on the steering wheel. In the suspended metre-rule demonstration, two equal forces were applied on opposite sides of the centre, pulling in opposite directions (anti-parallel), and the rule rotated. (a) What is a couple? (b) What is the net force on the rule from the two pulls, and what happens to the rule? Explain how this can be. (c) How do you calculate the moment of a

- (a) A couple is a pair of equal, parallel forces acting in opposite directions (anti-parallel) whose lines of action do not coincide.
- (b) The net force is zero because the two forces are equal and opposite, so the rule does not move in a straight line. It still rotates, because each force produces a moment in the same turning direction about the centre, so the moments add. A couple gives pure rotation with no resultant translational force.
- (c) Moment of a couple =  $F \times d$ , where F is the size of one of the forces and d is the perpendicular distance between the two forces.
- (d) Placing her hands further apart increases d, the perpendicular distance between the two forces. The force F from each hand stays the same, but  $F \times d$  becomes larger. This gives a greater turning effect on the steering wheel without extra pushing, so the wheel turns more easily.
- (e) Any two of: a pipe wrench used by a plumber in Kisumu, a tap wrench used by a plumber in Nairobi, the two-handed grip on a jembe handle.

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| couple? (d) Explain how a driver placing her hands further apart on the steering wheel helps her turn it, even though she does not push harder. (e) Give TWO other Kenyan examples of a couple. |  |
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| Part 5: Answering the driving question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <p><b>Driving question: Why does where you push, pull, or grip something determine how easy or hard it is to make it turn, and how do engineers and farmers in Kenya use this to design better tools?</b></p> <p><b>(a) Write a full answer of 4–6 sentences. Use the jembe in Kericho as your main example and include the terms pivot, perpendicular distance, force and moment. (b) Give TWO other examples from our school walk-around or video clips (such as a gate, tap, crane at Mombasa port, pliers at Kamukunji, or a boda-boda brake in Kisumu). For each, say where the pivot is and how the tool's design makes turning easier. (c) Compare your first prediction from Lesson 1 with your explanation now. What has changed in your thinking?</b></p> | <p>(a) A moment is the turning effect of a force, found by <math>M = F \times d</math>, where <math>d</math> is the perpendicular distance from the pivot to the line of action of the force. In the jembe, the blade-soil contact is the pivot. Where the farmer grips decides <math>d</math>. A longer handle, or gripping further from the blade, increases <math>d</math>, so the same effort produces a larger moment. That is why Amina, with the 90 cm handle, broke the soil more easily than Juma with the 30 cm handle, although the blade and the effort were the same. Engineers and farmers therefore design tools with long handles or wide grips to increase the turning effect without increasing the effort needed.</p> <p>(b) Example 1: the school gate. The pivot is the hinge. Pushing at the far edge, away from the hinge, gives a large <math>d</math>, so the gate swings open easily. Example 2: long-handled pliers used by a jua-kali artisan at Kamukunji. The pivot is the joint between the jaws. The long handles give a large effort arm, so a small squeeze produces a large moment on the jaws. (Other correct examples: the water tap handle with its pivot at the spindle, the boda-boda brake lever with its pivot at the lever bracket, or the crane at Mombasa port with its pivot at the base of the boom.)</p> <p>(c) Example answer: At first I only predicted that gripping further from the blade makes digging easier, without knowing why. Now I can explain it: the turning effect depends on both the force and the perpendicular distance from the pivot, <math>M = F \times d</math>, so a longer handle gives a larger moment for the same effort. I can also use the Principle of Moments and calculate the moments to prove it.</p> |

| RUBRIC                                               |                                                                                                                                                                       |                                                                                                                                        |                                                                                                         |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Criterion                                            | Excellent (4)                                                                                                                                                         | Proficient (3)                                                                                                                         | Developing (1–2)                                                                                        |
| Use of $M = F \times d$ and the Principle of Moments | Calculates every moment correctly with units, converts masses to weights using $g = 10 \text{ N/kg}$ , identifies clockwise and anticlockwise moments, and states the | Calculates most moments correctly with units and states the Principle of Moments with minor gaps, such as a missing direction or unit. | Makes frequent calculation errors, omits units or directions, or cannot state the Principle of Moments. |

|                                                        |                                                                                                                                                                                                                                             |                                                                                                                                                |                                                                                                                              |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                                                        | Principle of Moments accurately.                                                                                                                                                                                                            |                                                                                                                                                |                                                                                                                              |
| Explaining the Jembe Handle Puzzle                     | Identifies the blade-soil contact as the pivot and explains that the longer handle (90 cm against 30 cm) gives a larger perpendicular distance and so a larger moment for the same force, rejecting the idea that a bigger force is needed. | Explains that a longer handle makes digging easier and refers to distance and moment, but the pivot or the same-force reasoning is incomplete. | Says only that a longer handle is easier or gives vague reasons, without linking to moment, pivot or perpendicular distance. |
| Understanding of couples and net moment                | Defines a couple correctly, explains zero net force with pure rotation, uses $F \times d$ between the forces, and explains a real couple such as the two-handed steering wheel grip. Uses the perpendicular distance, not the arrow length. | Defines a couple and gives $F \times d$ , with small gaps in explaining net force or the steering wheel example.                               | Confuses a couple with a single force, or does not explain why the object turns.                                             |
| Connecting to Kenyan contexts and the driving question | Gives a coherent answer to the driving question with at least two correct Kenyan examples, each with pivot and design reason, and clearly shows how thinking changed since Lesson 1.                                                        | Gives a mostly complete answer with two examples, where the pivot or design reasoning is partly missing.                                       | Gives one example or none, with little link to the driving question or to the change in thinking.                            |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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